VENKATRAMAN RENGANATHAN



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IIT Hyderabad, Kandi, Telangana, 502284, India

Professional Experience

Assistant Professor - Grade I
Department of Artificial Intelligence, IIT Hyderabad, India *July 2026 - Present*

Lecturer in Multiple Agent Systems & Control
Faculty of Engineering & Applied Sciences, Cranfield University, UK *August 2024 - June 2026*

Visiting Assistant Professor (DAAD CPS AI-Net Fellow)
Control and Cyber-Physical Systems Laboratory, TU Darmstadt, Germany *November 2024*
Host: Dr. Rolf Findeisen

Postdoctoral Research Fellow
Department of Automatic Control, LTH - Lund University, Sweden *August 2021 - July 2024*
Mentor: Dr. Anders Rantzer

Graduate Research Assistant
Department of Mechanical Engineering - UT Dallas, USA *August 2016 - August 2021*
Advisor: Dr. Tyler Holt Summers

Software Engineer
Aspire Systems - India Private Limited, Chennai - India *June 2011 - September 2012*

Education

PhD, Mechanical Engineering (Dynamics & Controls), **CGPA: 3.95/4.0** *June 2021*
Thesis Advisor: Dr. Tyler Holt Summers
The University of Texas at Dallas, Richardson, TX, USA

Master of Science, Electrical Engineering (Control Systems), **CGPA: 3.76/4.0** *July 2016*
Thesis Supervisor: Dr. Armando Antonio Rodriguez
Arizona State University, Tempe, AZ, USA

Bachelor of Engineering, Electrical & Electronics Engineering, **CGPA: 9.12/10.0** *May 2011*
Project Advisor: Dr. V. Prasanna Moorthy
Government College of Technology - Anna University, Coimbatore, TN, India

Awards

- DAAD AI-Net for CPS Postdoc Short-term Research Visit Scholarship, October 2024.
- DAAD AI-Net for CPS Postdoc Networking Fellowship, October 2022.
- Travel Fellowship from Lund University - Faculty of Engineering LTH, 2022-2023.
- Louis Beecherl Jr. Fellowships from UT Dallas for the academic years 2019-2020 & 2020-2021.
- GCT Alumni Association Scholarship for Academic Excellency during the year 2010.

Journal Publications

1. **V. Renganathan**, S. Ghosh, "Probabilistic Assume–Guarantee Contracts for Islanded Mission-Critical Power System Operations", *MDPI Electronics*, 2026.
2. **V. Renganathan**, S. Safaoui, A. M. Kothari, B. Gravell, I. Shames, T. Summers, "Risk Bounded Nonlinear Motion Planning With Integrated Perception & Control", *Special Issue on "Risk-aware Autonomous Systems: Theory & Practice", Artificial Intelligence*, 2023.
3. **V. Renganathan**, B. Gravell, J.Ruths, T. Summers, "Anomaly Detection Under Multiplicative Noise Model Uncertainty", *IEEE Letters to Control Systems Society*, 2022.
4. **V. Renganathan**, N. Hashemi, J. Ruths, T. Summers, "Higher-Order Moment-Based Anomaly Detection", *IEEE Letters to Control Systems Society*, 2022.

5. **V. Renganathan**, K. Fathian, S. Safaoui, T. Summers, "Spoof resilient coordination in distributed & robust robotic networks", *IEEE Transaction on Control Systems Technology*, 2021.

Conference Publications

1. J. U. Mohammed, **V. Renganathan**, S. Mondal, "Automated Marine Plastic Monitoring", *13th International Conference on RITA*, 2025.
2. A. Navalakhe, S. Mondal, **V. Renganathan**, A. Tsourdos, " Optimising UAV Swarm Search Operations for Persistent Surveillance", *13th International Conference on RITA*, 2025.
3. A. Nyström, **V. Renganathan**, M. Cantoni, "Stereographic projection of probabilistic frequency-domain uncertainty", *Accepted for IEEE CDC*, 2024.
4. A. Banse, **V. Renganathan**, R. Jungers, "A Cantor-Kantorovich Metric Between Markov Decision Processes With Application to Transfer Learning", *MTNS*, Cambridge, UK, 2024.
5. **V. Renganathan**, A. Rantzer, O. Kjellqvist, "Distributed Minimax Adaptive Control For Uncertain Networks", *European Control Conference*, Stockholm, 2024.
6. M. Pfefferkorn, **V. Renganathan**, R. Findeisen, "Regret & Conservatism of Stochastic MPC", *Accepted for IEEE ACC*, 2024.
7. **V. Renganathan**, A. Iannelli, A. Rantzer, "Online Learning Analysis for Minimax Adaptive Control", *IEEE CDC*, Singapore, 2023.
8. C. Alptürk, **V. Renganathan**, "Risk Averse Path Planning Using Lipschitz Approximated Wasserstein Distributionally Robust Deep Q-learning", *ECC*, Bucharest, Romania, 2023.
9. **V. Renganathan, A. Cervin**, A. Rantzer, et.al, "Learning-based Control and Estimation for Attitude Regulation of a Reusable Launcher for Landing Scenario", *ESA-GNC & ICATT Conference*, Sopot, Poland, 2023.
10. **V. Renganathan**, J. Pilipovsky, P. Tsoitras, "Distributionally Robust Covariance Steering With Optimal Risk Allocation", *Accepted to IEEE ACC*, 2023.
11. K. Ekenberg, **V. Renganathan**, B. Olofsson, "Distributionally Robust RRT with Risk Allocation", *Accepted to IEEE ICRA*, 2023.
12. T. Jouini, Z. Sun, **V. Renganathan**, Veit Hagenmeyer, "Input and state constrained inverse optimal control with application to power networks", *IFAC World Congress*, 2023.
13. **V. Renganathan**, B. Gravell, J.Ruths, T. Summers, "Anomaly Detection Under Multiplicative Noise Model Uncertainty", *IEEE American Control Conference*, Atlanta, USA 2022.
14. S. Safaoui, B. Gravell, **V. Renganathan**, T. Summers, "Risk-Averse RRT* Planning with Nonlinear Steering and Tracking Controllers for Nonlinear Robotic Systems Under Uncertainty", *IEEE IROS*, 2021.
15. **V. Renganathan**, I. Shames, T. Summers, "Towards Integrated Perception and Motion Planning with Distributionally Robust Risk Constraints", *IFAC World Congress*, 2020.
16. **V. Renganathan**, N. Hashemi, J. Ruths, T. Summers, "Distributionally Robust Tuning of Anomaly Detectors in Cyber-Physical Systems with Stealthy Attacks", *IEEE American Control Conference*, 2020.
17. V. Raghuraman, **V. Renganathan**, T. Summers, J. Koeln, "Hierarchical MPC with Coordinating Terminal Costs", *IEEE American Control Conference*, 2020.
18. **V. Renganathan**, T. Summers, "Spoof Resilient Coordination for Distributed Multi-Robot Systems", *Proceedings of The International Symposium on Multi-Robot and Multi-Agent Systems (MRS)*, Los Angeles, USA, Dec 4-5, 2017.

Books & Theses

1. **V. Renganathan**, "Robust, resilient, and risk-aware optimization and controls for Cyber-physical systems", *PhD (ME) Thesis*, The University of Texas at Dallas, 2021.
2. **V. Renganathan**, "Kill Zone Analysis for a Bank-to-Turn Missile-Target Engagement", *MS(EE) Thesis*, Arizona State University, 2016.
3. **V. Renganathan et al.**, "PLC Logic Implementation for Controlling Tube-Bending Machine Operation", *BE(EEE) Thesis Project*, BHEL Trichy and Government College of Technology, Coimbatore, India, 2011.

Preprints Submitted & in Preparation

- **V. Renganathan**, "Probabilistic Robustness in the Gap Metric", *Manuscript Submitted & Under Review for IEEE TAC*, 2025.
- **V. Renganathan**, "Distance Between Stochastic Linear Systems", *Manuscript Submitted & Under Review for IEEE TAC*, 2025.
- **V. Renganathan**, "Chance Constrained Decoding for Risk Averse LLMs", *TMLR Journal Manuscript in Preparation*, 2026.
- D. Sundarsingh, **V. Renganathan**, Y. Kantaros, "Risk-Aware Path Planning for Swarms", *Journal Manuscript In Preparation for the International Journal of Robotics Research*, 2026.
- A. Navalakhe, **V. Renganathan**, S. Mondal, "Resilient Framework for Persistent Surveillance by UAV Swarm", *Manuscript in Preparation for IEEE Transactions on Aerospace & Electronics*, 2026.
- **V. Renganathan**, A. Nyström, M. Pfefferkorn, M. Cantoni, S.Z. Khong, "Probabilistic Robustness in the Pointwise-Gap Metric", *Manuscript in Preparation for IEEE TAC*, 2026.

Invited Talks

1. **Date:** December 18, 2025
   Title: Probabilistically Robust Decision Making for Uncertain Dynamical Systems
   Venue: Department of Artificial Intelligence, IIT Hyderabad, India.
2. **Date:** June 12, 2025
   Title: Probabilistic Robust Control Using ν-Gap Metric
   Venue: Control Group, University of Cambridge, United Kingdom
   Host: Dr. Fulvio Forni.
3. **Date:** June 04, 2025
   Title: Probabilistically Robust Decision Making for Uncertain Dynamical Systems
   Venue: Laboratory of Signals & Systems (L2S), CentraleSupélec, University of Paris-Saclay, France
   Host: Dr. Antoine Girard.
4. **Date:** February 16, 2023
   Title: Distributed Minimax Adaptive Controller for Finite Set of Linear Systems
   Venue: Department of Process Engineering, TU Berlin, Germany
   Host: Dr. Steffi Knorn.
5. **Date:** February 14, 2023
   Title: Distributionally Robust Covariance Steering with Optimal Risk Allocation
   Venue: Institute for Systems Theory & Automatic Control (IST), University of Stuttgart, Germany
   Host: Dr. Andrea Iannelli & Dr. Frank Allgower.
6. **Date:** February 13, 2023
   Title: Anomaly Detection for Cyber-physical Systems
   Venue: Institute of Industrial Automation and Software Engineering (IAS), University of Stuttgart, Germany
   Host: Dr. Andrey Morozov.

7. **Date:** February 07, 2023
   Title: Motion Planning Under Uncertainty With Risk Allocation
   Venue: Control & Cyber-Physical Systems Laboratory (CCPS), TU Darmstadt, Germany
   Host: Dr. Rolf Findeisen.
8. **Date:** January 18, 2023
   Title: Distributionally Robust Covariance Steering with Optimal Risk Allocation
   Venue: Department of Electrical Engineering, IIT Madras, India
   Host: Dr. Puduru Viswanadha Reddy.

Students Mentored

Role: PhD Co-Supervisor

1. **Abubakar Ibrahim** - Cranfield University, UK. (2025-2026) (Main Supervisor: D. Ignatyev)
   Thesis Topic: Sum of Squares for Region of Attraction Estimation with Parallel Computing & Multiple Shape Functions.
2. **Guillermo Zaragoza Prous** - Cranfield University, UK. (2025-2026) (Main Supervisor: L. Felicetti)
   Thesis Topic: Controlled Landing of Next Generation Re-entry Launch Vehicles.
3. **Olle Kjellqvist** - Lund University, Sweden. (2021-2024) (Main Supervisor: A. Rantzer)
   Thesis Topic: On Control and Estimation of Large and Uncertain Systems.

Role: Masters Thesis Supervisor

1. **Cem Alptürk** - Lund University, Sweden - Spring 2022.
   Thesis Topic: Risk Averse Path Planning Using Distributionally Robust Deep Reinforcement Learning. (Research Paper Accepted to ECC 2023).
2. **Kajsa Ekenberg** - Lund University, Sweden - Spring 2022.
   Thesis Topic: Distributionally Robust Risk Bounded Path Planning Through Iterative Spatio-temporal Risk Allocation. (Research Paper Accepted to ICRA 2023).
3. **Anton Nyström** - Lund University, Sweden - Spring 2024.
   Thesis Topic: Understanding Probabilistic Uncertainties Using ν-Gap. (Paper Accepted to IEEE CDC 2024 & Preparing Two Journal Manuscripts).
4. **Abhirva Navalakhe** - Cranfield University, UK - Summer 2025.
   Thesis Topic: Optimizing Swarm Search Using Convex Optimization. (Paper Accepted to International Conference on RITA 2025, London, UK & Preparing a Journal Manuscript)
5. **Rohit Bhargava** - Cranfield University, UK - Summer 2025.
   Thesis Topic: Obstacle Detection Using Event-Based Camera. (Paper Accepted to International Conference on RITA 2025, London, UK).
6. **Jamal Mohammed** - Cranfield University, UK - Summer 2025.
   Thesis Topic: Machine Learning Approaches for Ocean Plastic Monitoring. (Paper Accepted to International Conference on RITA 2025, London, UK).

Role: Masters Co-Supervisor

1. **Yi Gou** - Cranfield University, UK. (2025-Present) (Main Supervisor: S. Mondal)
   Thesis Topic: Design of Cooperative Autonomous Vehicle Behavior Algorithms in Communication & Bandwidth Limited Environments. (Paper Accepted to International Conference on RITA 2025, London, UK).

Role: Mentor

1. **Aadi M Kothari** - Bachelor Student Researcher at UT Dallas, USA (2020-2021).
   Current Position: Completed MS in Mechanical Engineering at MIT, Boston, USA.

Reviewing Activities

Journals

- IFAC Automatica
- IEEE Transaction on Automatic Control
- IEEE Transaction on Control of Network Systems
- IEEE Letters to Control Systems Society
- IEEE Transaction on Robotics
- Systems & Controls Letters
- Control Engineering Practice
- IEEE Signal Processing Letters
- IEEE Robotics & Automation Magazine

Conferences

- IEEE CDC, ACC, CCTA
- IFAC World Congress
- L4DC
- IROS, ICRA
- ASME DSCC

Sponsored Research & Consulting Activities

- **European Space Agency (ESA):** I was part of the ESA sponsored *Artificial Intelligence for Guidance Navigation & Control (AI4GNC)* research consortium representing Lund University from 2021 till 2023.

Other Activities

Session Co-chair at IEEE CDC 2022
I served as the Session Co-chair for a technical session at IEEE Conference on Decision & Control (CDC) Conference that was held during December 2022 at Cancun, Mexico.

Important Service Roles

1. **Masters Application Evaluator - Cranfield University, UK:** I helped the selection committee in selecting Masters students from a pool of around 50 applicants for the MSc program on Autonomous Vehicle Design & Control during 2025 and 2026.
2. **PhD Application Evaluator - Lund University, Sweden:** I was responsible for selecting PhD students from a pool of around 90 applicants for filling multiple PhD positions at the department of automatic control in Lund University, Sweden during 2022.

Teaching Experience

Department of Artificial Intelligence, IIT Hyderabad, India
Role: Course Instructor
Multi-Agent Systems — Fall 2026

Faculty of Engineering & Applied Sciences, Cranfield University, UK
Role: Module Leader & Module Instructor
Multi-Agent Systems — Spring 2026

Role: Faculty Supervisor - Group Design Project

Aircraft Inspection in Hangar Facility Using UAV Swarms — Oct 2025 - Mar 2026
CraneAero - Autonomous Emergency Response UAV Platform — Oct 2025 - Mar 2026

Department of Automatic Control, Lund University, Lund, Sweden
Role: Course Instructor
PhD Course on Linear Systems — Spring 2022

Role: Teaching Assistant
Course on Learning Based Control — Spring 2022

School of Mechanical Engineering, UT Dallas - TX, USA
Role: Graduate Teaching Assistant
MECH 4310 - Systems and Controls — Spring 2020

Department of Mathematics & Statistics - ASU, AZ - USA
Role: Grader
MAT 265 - Calculus for Engineers - I — Fall 2013
MAT 267 - Calculus for Engineers - III — Spring 2014
MAT 265 - Elementary Linear Algebra — Summer 2014
MAT 210 - Brief Calculus — Fall 2014
MAT 265 - Calculus for Engineers - I — Spring 2015
MAT 265 - Calculus for Engineers - I — Spring 2016
MAT 211 - Math for Business Analysis — Spring 2016
MAT 210 - Brief Calculus — Spring 2016

Role: Instructional Aide
MAT 265 - Calculus for Engineers - I — Summer 2014
MAT 170 - Pre-Calculus — Summer 2014
MAT 142 - College Mathematics — Summer 2015
MAT 170 - Pre-Calculus — Fall 2015
MAT 142 - College Mathematics — Summer 2016

School of Mechanical, Aerospace, Chemical & Materials Engg., ASU - AZ, USA
Role: Grader for Dr. Daniel Rivera
CHE 461 - Process Dynamics and Controls — Fall 2014
CHE 494/598 - System Identification — Spring 2015

Personal Details

- **ORCID ID:** 0000-0003-1420-2140
- **Scopus ID:** 57215295169
- **Github:** https://github.com/venkatramanrenganathan
- **Webpage:** https://venkatramanrenganathan.github.io/

- **Nationality:** Indian
- **Languages:** Tamil, English, Hindi
- **Family Details:** Wife: Reshma (Married: 2019), Son: Vishruth (Born: 2022)